User: Task : section wise notes with visualization of html content (preview). Structure: explain the concept then state it's uses. E.g., when explaining html, you start explaining texts, and then in that section state all the related functions of css and other things inside it. According to: syllabus.

SECTION-WISE NOTES: FRONT END WEB DESIGN & DEVELOPMENT

Based on Delhi University Syllabus

UNIT 1: INTRODUCTION TO WEB TECHNOLOGIES

(Weeks 1-3: Internet, Web Design, Web Architecture)

1.1 Concept: Internet & Web Architecture

What it is: The internet is a global network of interconnected computers. Web architecture refers to how web applications are structured: Client (browser) → Request → Server → Response → Client.

Uses: - Understanding how websites load - Debugging connection issues - Optimizing performance - Security considerations

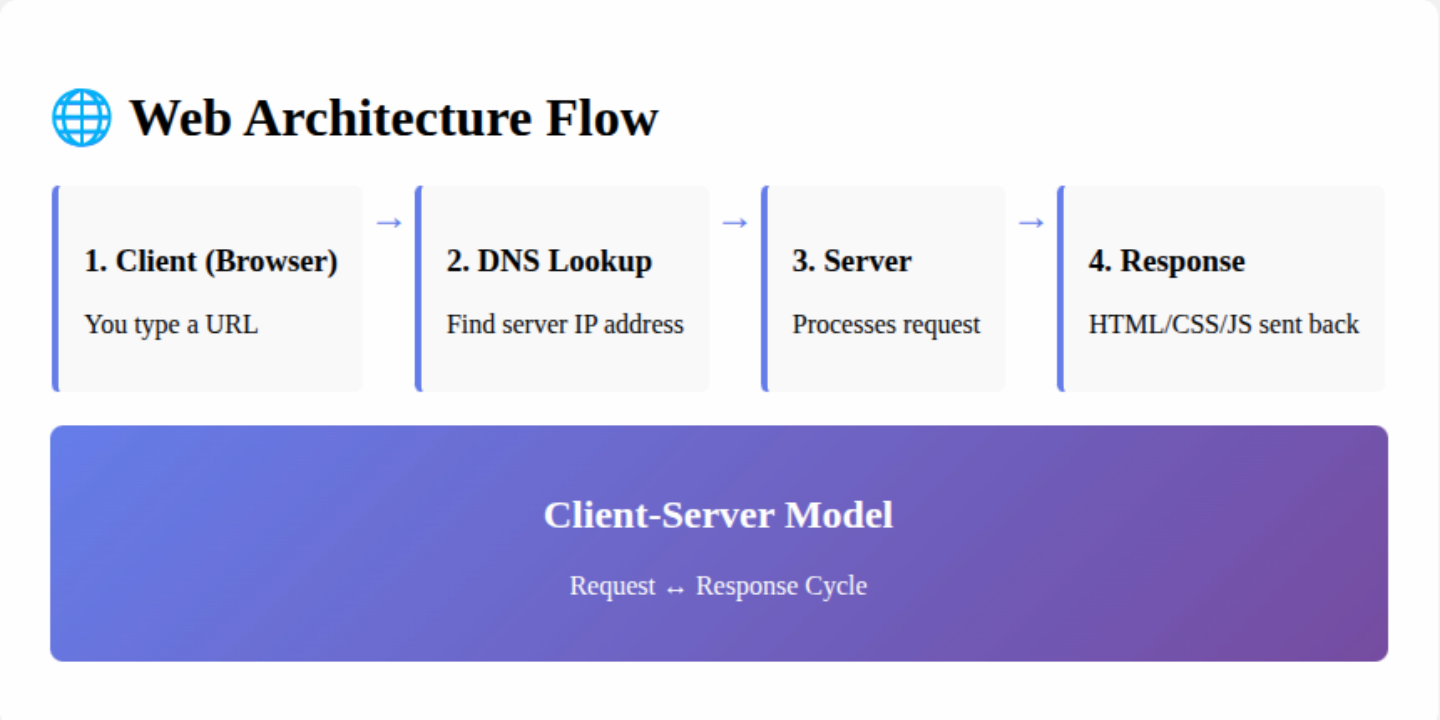
Key Components:

Component	Function	Example
Client	Browser that requests content	Chrome, Firefox
Server	Computer that hosts websites	Apache, Nginx
HTTP/HTTPS	Protocol for data transfer	`https://www.example.com`
DNS	Converts domain names to IP addresses	`google.com` → `142.250.80.46`

Visualization:

```html
<!DOCTYPE html>
<html lang="en">
<head>
 <meta charset="UTF-8">
 <title>Web Architecture Demo</title>
 <style>
 body { font-family: Arial; background: #f0f0f0; padding: 20px; }
 .container { max-width: 800px; margin: 0 auto; background: white; padding: 30px; border-radius: 10px; }
 .architecture-box {
 background: linear-gradient(135deg, #667eea 0%, #764ba2 100%);
 color: white;
 padding: 20px;
 margin: 10px 0;
 border-radius: 8px;
 text-align: center;
```

```
      }
      .arrow { text-align: center; font-size: 24px; color: #667eea; margin: 5px 0; }
      .flow { display: flex; justify-content: space-around; margin: 20px 0; }
      .step { background: #f8f9fa; padding: 15px; border-radius: 5px; border-left: 4px solid #667eea; }
    </style>
  </head>
  <body>
    <div class="container">
      <h1>     Web Architecture Flow</h1>

      <div class="flow">
        <div class="step">
          <h3>1. Client (Browser)</h3>
          <p>You type a URL</p>
        </div>
        <div class="arrow">→</div>
        <div class="step">
          <h3>2. DNS Lookup</h3>
          <p>Find server IP address</p>
        </div>
        <div class="arrow">→</div>
        <div class="step">
          <h3>3. Server</h3>
          <p>Processes request</p>
        </div>
        <div class="arrow">→</div>
        <div class="step">
          <h3>4. Response</h3>
          <p>HTML/CSS/JS sent back</p>
        </div>
      </div>

      <div class="architecture-box">
        <h2>Client-Server Model</h2>
        <p>Request      Response Cycle</p>
      </div>
    </div>
  </body>
  </html>
```



UNIT 2: HTML (HYPERTEXT MARKUP LANGUAGE)

(Weeks 4-7: Creating web pages, lists, hyperlinks, tables, forms, images, frames)

2.1 Concept: Text Formatting & Basic Structure

What it is: HTML uses tags to structure content. Text formatting tags define how content appears and its semantic meaning.

Uses: - **SEO:** Search engines read headings (`<h1>`) to understand content hierarchy - **Accessibility:** Screen readers use semantic tags for navigation - **Maintainability:** Proper structure makes code easier to update

Related CSS Integration:

HTML Element	CSS Property	Purpose
`<h1>` - `<h6>`	`font-size`, `color`, `text-align`	Visual hierarchy
`<p>`	`line-height`, `margin`, `text-indent`	Readability
`<strong>`	`font-weight: bold`	Emphasis
`<em>`	`font-style: italic`	Stress emphasis
`<u>`	`text-decoration: underline`	Annotation

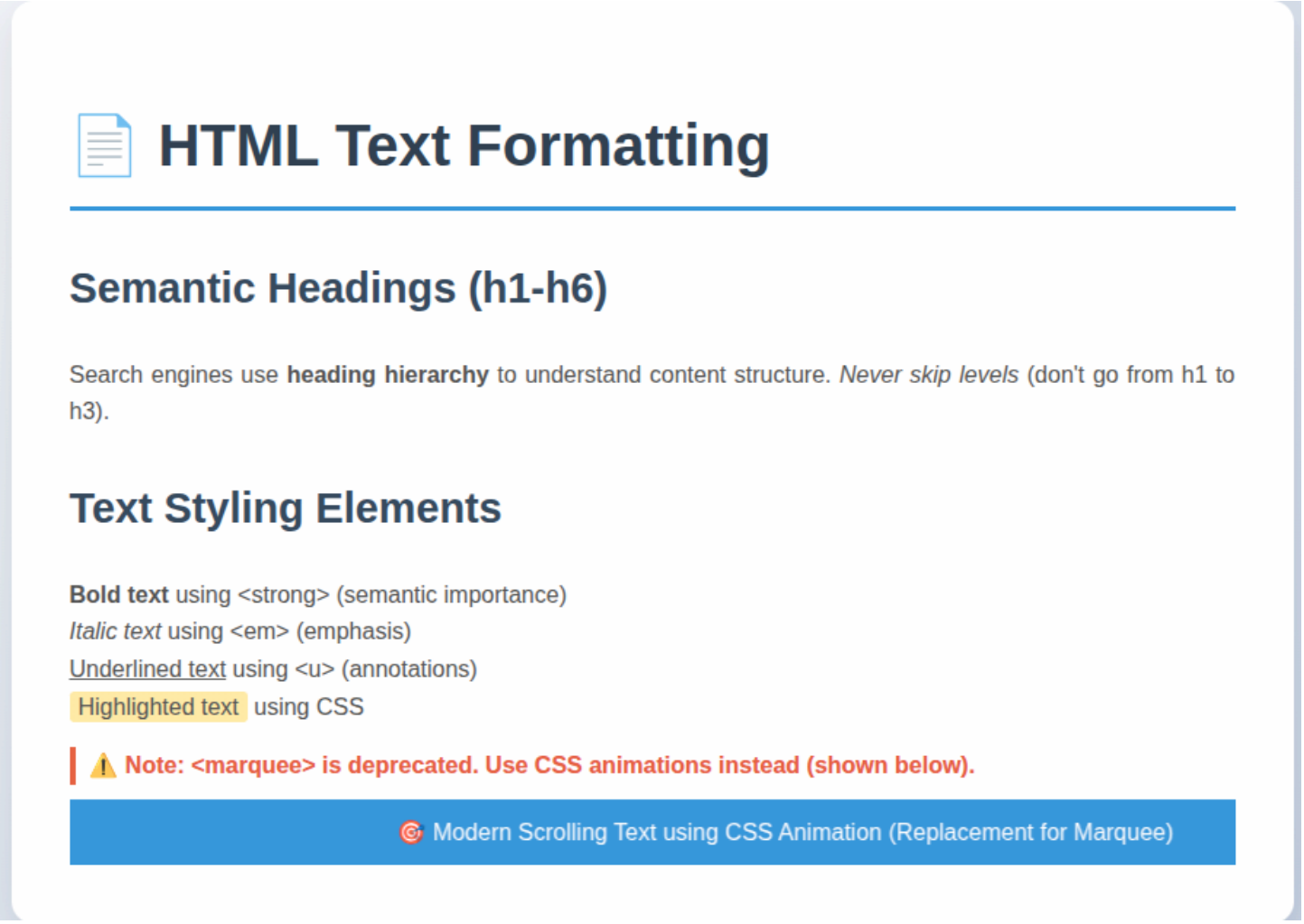
Visualization:

```
<!DOCTYPE html>
<html lang="en">
<head>
 <meta charset="UTF-8">
```

```
<title>Text Formatting & Structure</title>
<style>
    body {
        font-family: 'Segoe UI', sans-serif;
        background: linear-gradient(135deg, #f5f7fa 0%, #c3cfe2 100%);
        padding: 40px;
        line-height: 1.6;
    }
    .content-box {
        max-width: 800px;
        margin: 0 auto;
        background: white;
        padding: 40px;
        border-radius: 15px;
        box-shadow: 0 10px 30px rgba(0,0,0,0.1);
    }

    /* Typography Hierarchy */
    h1 {
        color: #2c3e50;
        border-bottom: 3px solid #3498db;
        padding-bottom: 10px;
        font-size: 2.5em;
    }
    h2 {
        color: #34495e;
        margin-top: 30px;
        font-size: 1.8em;
    }

    /* Text Formatting Classes */
    .highlight {
        background: linear-gradient(120deg, #ffeaa7 0%, #ffeaa7 100%);
        padding: 2px 6px;
        border-radius: 3px;
    }
    .important {
        color: #e74c3c;
        font-weight: bold;
        border-left: 4px solid #e74c3c;
        padding-left: 10px;
        display: block;
        margin: 10px 0;
    }

    /* Paragraph Styling */
    p {
        color: #555;
```

```
        text-align: justify;
        margin-bottom: 15px;
    }

    /* Marquee Replacement with CSS Animation */
    .scrolling-text {
        background: #3498db;
        color: white;
        padding: 10px;
        overflow: hidden;
        white-space: nowrap;
        position: relative;
    }
    .scrolling-text span {
        display: inline-block;
        animation: scroll 10s linear infinite;
    }
    @keyframes scroll {
        0% { transform: translateX(100%); }
        100% { transform: translateX(-100%); }
    }
  </style>
</head>
<body>
  <div class="content-box">
    <h1>     HTML Text Formatting</h1>

    <h2>Semantic Headings (h1-h6)</h2>
    <p>Search engines use <strong>heading hierarchy</strong> to understand content structure.
    <em>Never skip levels</em> (don't go from h1 to h3).</p>

    <h2>Text Styling Elements</h2>
    <p>
        <strong>Bold text</strong> using &lt;strong&gt; (semantic importance)<br>
        <em>Italic text</em> using &lt;em&gt; (emphasis)<br>
        <u>Underlined text</u> using &lt;u&gt; (annotations)<br>
        <span class="highlight">Highlighted text</span> using CSS
    </p>

    <div class="important">
            Note: &lt;marquee&gt; is deprecated. Use CSS animations instead (shown below).
    </div>

    <div class="scrolling-text">
        <span>     Modern Scrolling Text using CSS Animation (Replacement for Marquee)</span>
    </div>
  </div>
```

```
</body>
</html>
```



2.2 Concept: Lists (Ordered & Unordered)

What it is: Lists organize content into readable, structured formats. `<ul>` for bullet points, `<ol>` for numbered sequences.

Uses: - **Navigation menus:** Most website navs are styled lists - **Procedures:** Step-by-step instructions - **Data display:** Product features, ingredients

Related CSS Integration:

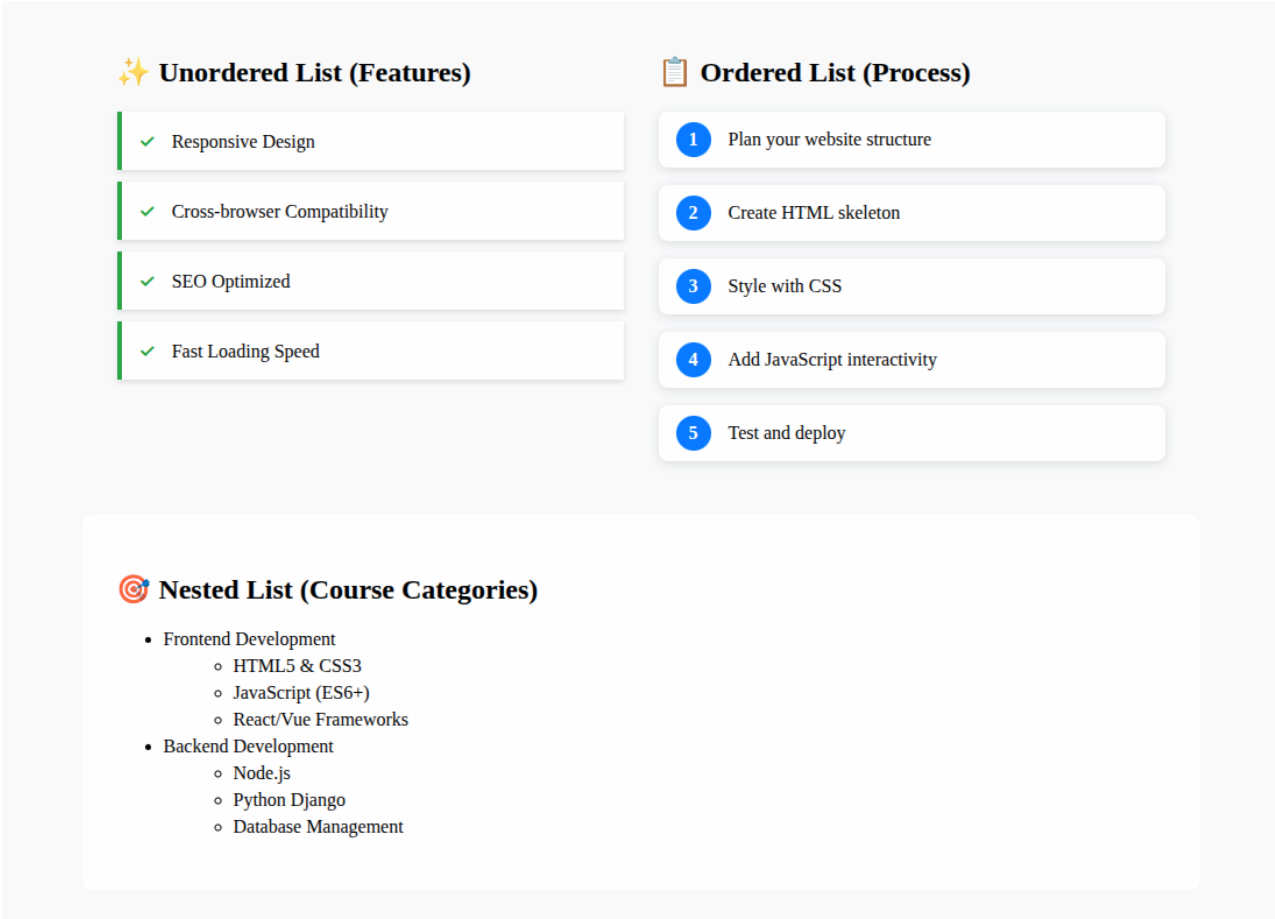
CSS Property	Effect
`list-style-type`	Change bullet style (disc, circle, square, decimal, roman)
`list-style-position`	Inside or outside text flow
`list-style-image`	Custom bullet images

Visualization:

```html
<!DOCTYPE html>
<html lang="en">
<head>
 <meta charset="UTF-8">
 <title>Lists & Navigation</title>
 <style>
 body { font-family: Arial; background: #f8f9fa; padding: 30px; }
 .container { max-width: 900px; margin: 0 auto; display: grid; grid-template-columns: 1fr 1fr; gap:
30px; }

 /* Unordered List Styling */
 .feature-list {
 list-style: none;
 padding: 0;
 }
 .feature-list li {
 background: white;
 margin: 10px 0;
 padding: 15px;
 border-left: 4px solid #28a745;
 box-shadow: 0 2px 5px rgba(0,0,0,0.1);
 transition: transform 0.2s;
 }
 .feature-list li:hover {
 transform: translateX(10px);
 background: #f8f9fa;
 }
 .feature-list li::before {
 content: "✓";
 color: #28a745;
 font-weight: bold;
 margin-right: 10px;
 }

 /* Ordered List Styling */
 .step-list {
 counter-reset: step-counter;
 list-style: none;
 padding: 0;
 }
 .step-list li {
 counter-increment: step-counter;
 margin: 15px 0;
 padding: 15px 15px 15px 60px;
 background: white;
 border-radius: 8px;
```

```
        position: relative;
        box-shadow: 0 2px 8px rgba(0,0,0,0.1);
    }
    .step-list li::before {
        content: counter(step-counter);
        position: absolute;
        left: 15px;
        top: 50%;
        transform: translateY(-50%);
        background: #007bff;
        color: white;
        width: 30px;
        height: 30px;
        border-radius: 50%;
        display: flex;
        align-items: center;
        justify-content: center;
        font-weight: bold;
    }

    /* Nested List */
    .nested-list li { margin: 5px 0; }
    .nested-list ul { margin-top: 5px; margin-left: 20px; }
  </style>
</head>
<body>
  <div class="container">
    <div>
      <h2>     Unordered List (Features)</h2>
      <ul class="feature-list">
        <li>Responsive Design</li>
        <li>Cross-browser Compatibility</li>
        <li>SEO Optimized</li>
        <li>Fast Loading Speed</li>
      </ul>
    </div>

    <div>
      <h2>     Ordered List (Process)</h2>
      <ol class="step-list">
        <li>Plan your website structure</li>
        <li>Create HTML skeleton</li>
        <li>Style with CSS</li>
        <li>Add JavaScript interactivity</li>
        <li>Test and deploy</li>
      </ol>
    </div>
  </div>
```

```
  <div style="max-width: 900px; margin: 30px auto; background: white; padding: 30px; border-radius: 10px;">
    <h2>    Nested List (Course Categories)</h2>
    <ul class="nested-list">
      <li>Frontend Development
        <ul>
          <li>HTML5 & CSS3</li>
          <li>JavaScript (ES6+)</li>
          <li>React/Vue Frameworks</li>
        </ul>
      </li>
      <li>Backend Development
        <ul>
          <li>Node.js</li>
          <li>Python Django</li>
          <li>Database Management</li>
        </ul>
      </li>
    </ul>
  </div>
</body>
</html>
```



2.3 Concept: Hyperlinks (Internal & External)

What it is: The <a> (anchor) tag creates clickable links to other pages, sections, or external websites.

Uses: - **Navigation:** Moving between pages - **Table of contents:** Jumping to sections within long pages - **References:** Citing external sources - **Downloads:** Linking to files

Related CSS Integration:

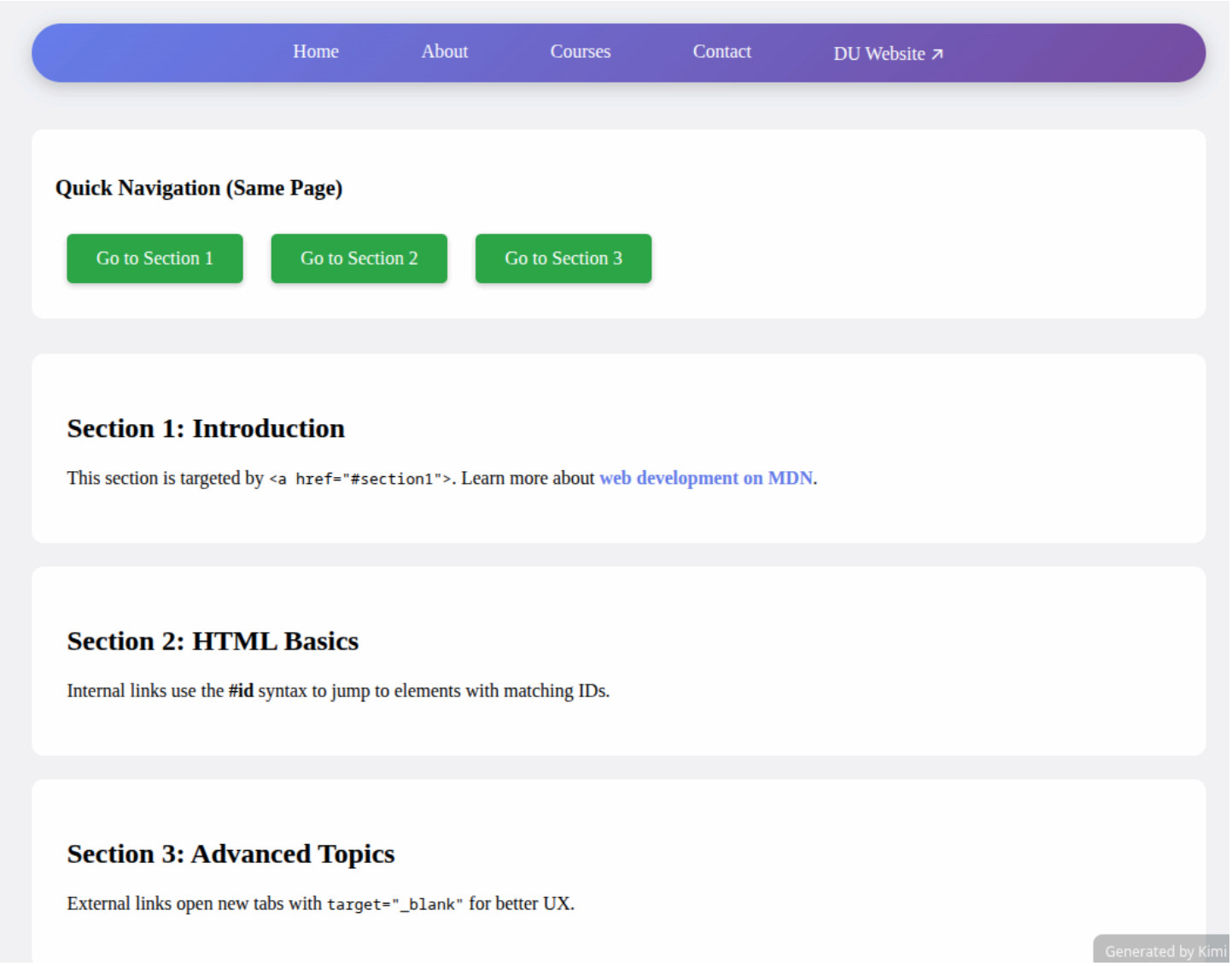
State	CSS Pseudo-class	Styling Purpose
Unvisited	`a:link`	Default state
Visited	`a:visited`	Show user has been there
Hover	`a:hover`	Mouse over feedback
Active	`a:active`	Click moment

Visualization:

```
<!DOCTYPE html>
<html lang="en">
<head>
 <meta charset="UTF-8">
 <title>Links & Navigation</title>
 <style>
 body { font-family: 'Segoe UI'; background: #f0f2f5; padding: 40px; }
 .nav-container { max-width: 1000px; margin: 0 auto; }

 /* Navigation Bar */
 .main-nav {
 background: linear-gradient(135deg, #667eea 0%, #764ba2 100%);
 padding: 15px;
 border-radius: 50px;
 margin-bottom: 40px;
 box-shadow: 0 4px 15px rgba(0,0,0,0.2);
 }
 .main-nav ul {
 list-style: none;
 display: flex;
 justify-content: center;
 gap: 30px;
 margin: 0;
 padding: 0;
 }
 .main-nav a {
 color: white;
 text-decoration: none;
 padding: 10px 20px;
 border-radius: 25px;
 transition: all 0.3s;
```

```
        font-weight: 500;
    }
    .main-nav a:hover {
        background: rgba(255,255,255,0.2);
        transform: translateY(-2px);
    }

    /* Content Links */
    .content-link {
        color: #667eea;
        text-decoration: none;
        border-bottom: 2px solid transparent;
        transition: border-color 0.3s;
        font-weight: 600;
    }
    .content-link:hover {
        border-bottom-color: #667eea;
    }

    /* Button-style Link */
    .btn-link {
        display: inline-block;
        background: #28a745;
        color: white;
        padding: 12px 25px;
        text-decoration: none;
        border-radius: 5px;
        margin: 10px;
        transition: all 0.3s;
        box-shadow: 0 2px 5px rgba(0,0,0,0.2);
    }
    .btn-link:hover {
        background: #218838;
        transform: translateY(-2px);
        box-shadow: 0 4px 8px rgba(0,0,0,0.3);
    }

    /* Anchor Links Section */
    .section {
        background: white;
        padding: 30px;
        margin: 20px 0;
        border-radius: 10px;
        scroll-margin-top: 100px;
    }
  </style>
</head>
<body>
```

```
  <div class="nav-container">
    <!-- External Navigation -->
    <nav class="main-nav">
      <ul>
        <li><a href="#home">Home</a></li>
        <li><a href="#about">About</a></li>
        <li><a href="#courses">Courses</a></li>
        <li><a href="#contact">Contact</a></li>
        <li><a href="https://www.du.ac.in" target="_blank">DU Website     </a></li>
      </ul>
    </nav>

    <!-- Internal Anchor Links -->
    <div style="background: white; padding: 20px; border-radius: 10px; margin-bottom: 30px;">
      <h3>Quick Navigation (Same Page)</h3>
      <a href="#section1" class="btn-link">Go to Section 1</a>
      <a href="#section2" class="btn-link">Go to Section 2</a>
      <a href="#section3" class="btn-link">Go to Section 3</a>
    </div>

    <!-- Content Sections -->
    <div class="section" id="section1">
      <h2>Section 1: Introduction</h2>
      <p>This section is targeted by <code>&lt;a href="#section1"&gt;</code>.
      Learn more about <a href="https://developer.mozilla.org" target="_blank" class="content-
link">web development on MDN</a>.</p>
    </div>

    <div class="section" id="section2">
      <h2>Section 2: HTML Basics</h2>
      <p>Internal links use the <strong>#id</strong> syntax to jump to elements with matching
IDs.</p>
    </div>

    <div class="section" id="section3">
      <h2>Section 3: Advanced Topics</h2>
      <p>External links open new tabs with <code>target="_blank"</code> for better UX.</p>
    </div>
  </div>
</body>
</html>
```



2.4 Concept: Tables

What it is: Tables (<table>, <tr>, <td>, <th>) display tabular data in rows and columns.

Uses: - **Timetables:** Class schedules (Syllabus Practical 3) - **Data comparison:** Pricing tables, feature lists - **Financial data:** Reports, statements - **Sports statistics:** Scores, standings

Related CSS Integration:

CSS Property	Effect
`border-collapse`	Remove double borders
`nth-child(even/odd)`	Zebra striping
`text-align`	Center/left/right align content
`vertical-align`	Top/middle/bottom alignment

Visualization:

```html
<!DOCTYPE html>
<html lang="en">
<head>
 <meta charset="UTF-8">
 <title>Tables & Timetables</title>
 <style>
 body { font-family: 'Segoe UI'; background: #f5f7fa; padding: 40px; }
 .container { max-width: 1000px; margin: 0 auto; }

 /* Basic Table Styling */
 .data-table {
 width: 100%;
 border-collapse: collapse;
 background: white;
 box-shadow: 0 4px 6px rgba(0,0,0,0.1);
 border-radius: 8px;
 overflow: hidden;
 }
 .data-table th {
 background: linear-gradient(135deg, #667eea 0%, #764ba2 100%);
 color: white;
 padding: 15px;
 text-align: left;
 font-weight: 600;
 }
 .data-table td {
 padding: 12px 15px;
 border-bottom: 1px solid #e0e0e0;
 }
 .data-table tr:hover {
 background-color: #f5f7fa;
 }
 .data-table tr:nth-child(even) {
 background-color: #fafafa;
 }

 /* Timetable Specific */
 .timetable {
 width: 100%;
 border-collapse: collapse;
 font-size: 14px;
 box-shadow: 0 4px 15px rgba(0,0,0,0.1);
 }
 .timetable th {
 background: #2c3e50;
 color: white;
```

```
        padding: 12px;
        text-align: center;
    }
    .timetable td {
        border: 1px solid #ddd;
        padding: 10px;
        text-align: center;
        height: 60px;
    }
    .timetable tr:nth-child(even) {
        background: #f8f9fa;
    }
    .lecture { background: #e3f2fd; color: #1976d2; font-weight: bold; }
    .lab { background: #f3e5f5; color: #7b1fa2; font-weight: bold; }
    .break { background: #fff3e0; color: #f57c00; }
  </style>
</head>
<body>
  <div class="container">
    <h2>     Student Information Table</h2>
    <table class="data-table">
      <thead>
        <tr>
          <th>Roll No</th>
          <th>Name</th>
          <th>Course</th>
          <th>Semester</th>
          <th>Grade</th>
        </tr>
      </thead>
      <tbody>
        <tr>
          <td>2024001</td>
          <td>Rahul Sharma</td>
          <td>B.Sc. CS</td>
          <td>4th</td>
          <td>A</td>
        </tr>
        <tr>
          <td>2024002</td>
          <td>Priya Patel</td>
          <td>B.A. Economics</td>
          <td>2nd</td>
          <td>A+</td>
        </tr>
        <tr>
          <td>2024003</td>
          <td>Amit Kumar</td>
```

```
            <td>B.Sc. Physics</td>
            <td>6th</td>
            <td>B+</td>
          </tr>
        </tbody>
      </table>

      <h2 style="margin-top: 40px;">     Class Timetable (Syllabus Practical 3)</h2>
      <table class="timetable">
        <thead>
          <tr>
            <th>Time</th>
            <th>Monday</th>
            <th>Tuesday</th>
            <th>Wednesday</th>
            <th>Thursday</th>
            <th>Friday</th>
          </tr>
        </thead>
        <tbody>
          <tr>
            <td><strong>9:00-10:00</strong></td>
            <td class="lecture">HTML/CSS</td>
            <td class="lab">Web Lab</td>
            <td class="lecture">JavaScript</td>
            <td class="lecture">Database</td>
            <td class="lab">Project Work</td>
          </tr>
          <tr>
            <td><strong>10:00-11:00</strong></td>
            <td class="lecture">JavaScript</td>
            <td class="lecture">HTML/CSS</td>
            <td class="lab">Web Lab</td>
            <td class="lecture">Networking</td>
            <td class="lecture">Database</td>
          </tr>
          <tr>
            <td><strong>11:00-12:00</strong></td>
            <td colspan="5" class="break">     BREAK</td>
          </tr>
        </tbody>
      </table>
    </div>
  </body>
</html>
```

📊 Student Information Table

Roll No	Name	Course	Semester	Grade
2024001	Rahul Sharma	B.Sc. CS	4th	A
2024002	Priya Patel	B.A. Economics	2nd	A+
2024003	Amit Kumar	B.Sc. Physics	6th	B+

🗓️ Class Timetable (Syllabus Practical 3)

Time	Monday	Tuesday	Wednesday	Thursday	Friday
9:00-10:00	HTML/CSS	Web Lab	JavaScript	Database	Project Work
10:00-11:00	JavaScript	HTML/CSS	Web Lab	Networking	Database
11:00-12:00	☕ BREAK				

2.5 Concept: Images & Media

What it is: The <img> tag embeds images. Attributes control display: src (source), alt (alternative text), width/height (dimensions).

Uses: - **Content illustration:** Photos, diagrams - **UI elements:** Icons, logos, buttons - **Accessibility:** Alt text for screen readers - **SEO:** Image search optimization

Related CSS Integration:

CSS Property	Effect
`max-width: 100%`	Responsive scaling
`border-radius`	Rounded corners
`box-shadow`	Depth effect
`object-fit`	Control how image fills container
`filter`	Visual effects (grayscale, brightness)

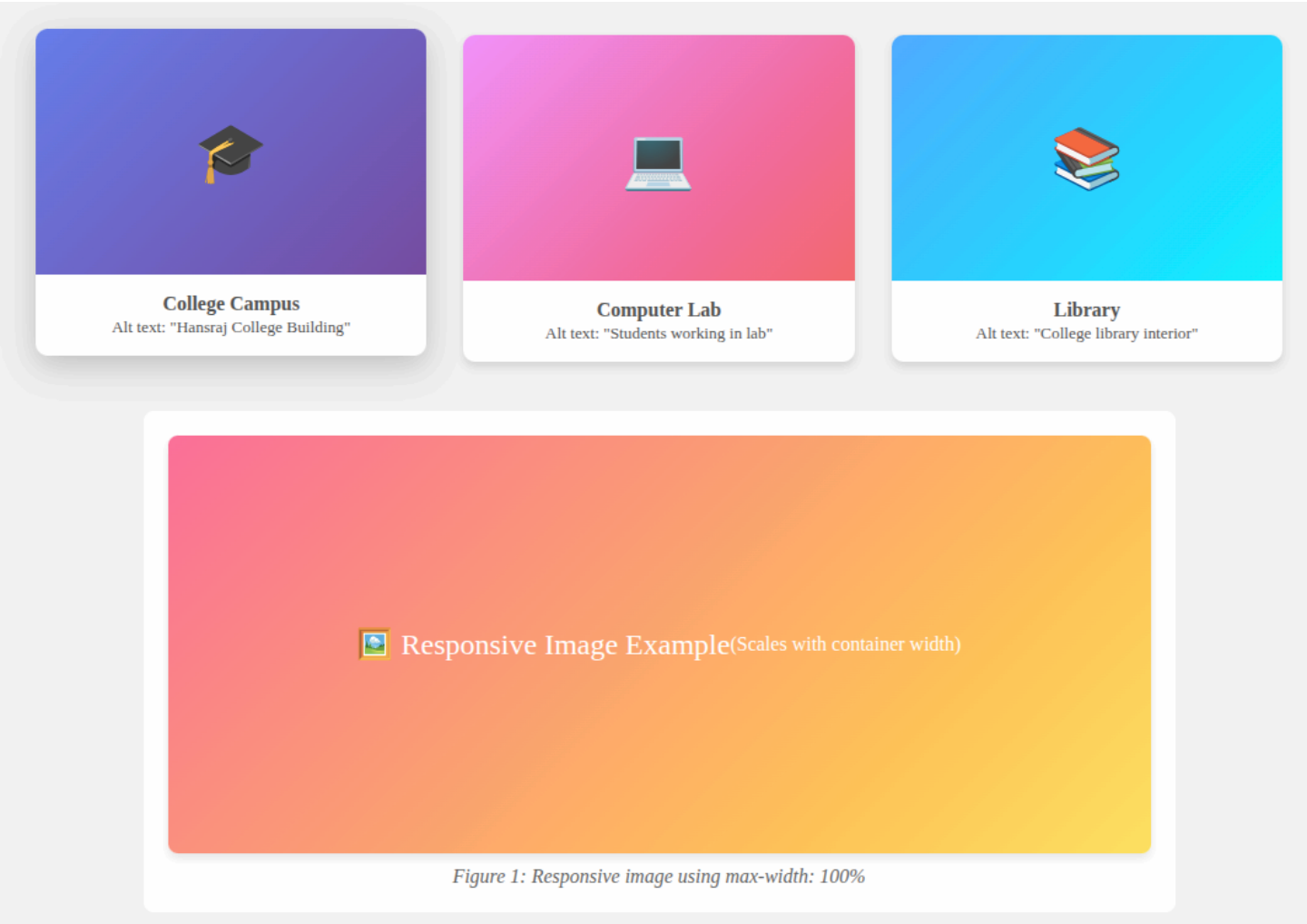
Visualization:

```html
<!DOCTYPE html>
<html lang="en">
<head>
 <meta charset="UTF-8">
```

```
<title>Images & Media</title>
<style>
  body { font-family: Arial; background: #f0f0f0; padding: 40px; }
  .gallery {
    display: grid;
    grid-template-columns: repeat(auto-fit, minmax(300px, 1fr));
    gap: 30px;
    max-width: 1200px;
    margin: 0 auto;
  }

  /* Image Card */
  .image-card {
    background: white;
    border-radius: 10px;
    overflow: hidden;
    box-shadow: 0 4px 6px rgba(0,0,0,0.1);
    transition: transform 0.3s;
  }
  .image-card:hover {
    transform: translateY(-5px);
    box-shadow: 0 8px 25px rgba(0,0,0,0.2);
  }
  .image-card img {
    width: 100%;
    height: 200px;
    object-fit: cover;
    display: block;
  }
  .image-caption {
    padding: 15px;
    text-align: center;
    color: #555;
  }

  /* Responsive Image */
  .responsive-img {
    max-width: 100%;
    height: auto;
    border-radius: 8px;
    box-shadow: 0 4px 8px rgba(0,0,0,0.1);
  }

  /* Figure Styling */
  figure {
    background: white;
    padding: 20px;
    border-radius: 8px;
```

```
        margin: 20px 0;
        text-align: center;
    }
    figcaption {
        margin-top: 10px;
        color: #666;
        font-style: italic;
    }
  </style>
</head>
<body>
  <div class="gallery">
    <!-- Image with Attributes -->
    <div class="image-card">
        <!-- Placeholder using CSS gradient -->
        <div style="width: 100%; height: 200px; background: linear-gradient(135deg, #667eea 0%,
#764ba2 100%); display: flex; align-items: center; justify-content: center; color: white; font-size: 48px;">

        </div>
        <div class="image-caption">
            <strong>College Campus</strong><br>
            <small>Alt text: "Hansraj College Building"</small>
        </div>
    </div>

    <div class="image-card">
        <div style="width: 100%; height: 200px; background: linear-gradient(135deg, #f093fb 0%,
#f5576c 100%); display: flex; align-items: center; justify-content: center; color: white; font-size: 48px;">

        </div>
        <div class="image-caption">
            <strong>Computer Lab</strong><br>
            <small>Alt text: "Students working in lab"</small>
        </div>
    </div>

    <div class="image-card">
        <div style="width: 100%; height: 200px; background: linear-gradient(135deg, #4facfe 0%, #00f2fe
100%); display: flex; align-items: center; justify-content: center; color: white; font-size: 48px;">

        </div>
        <div class="image-caption">
            <strong>Library</strong><br>
            <small>Alt text: "College library interior"</small>
        </div>
    </div>
  </div>
```

```
  <figure style="max-width: 800px; margin: 40px auto;">
    <div class="responsive-img" style="height: 300px; background: linear-gradient(135deg, #fa709a 0%, #fee140 100%); display: flex; align-items: center; justify-content: center; color: white; font-size: 24px; text-align: center; padding: 20px;">
            Responsive Image Example<br>
        <small style="font-size: 16px;">(Scales with container width)</small>
    </div>
    <figcaption>Figure 1: Responsive image using max-width: 100%</figcaption>
  </figure>
</body>
</html>
```



2.6 Concept: Web Forms

What it is: Forms (`<form>`, `<input>`, `<textarea>`, `<select>`) collect user input. The foundation of user interaction on the web.

Uses: - **User registration:** Creating accounts - **Data collection:** Surveys, feedback - **Search:** Query input - **E-commerce:** Checkout processes

Related CSS Integration:

Element	CSS Styling
`input[type="text"]`	`border`, `padding`, `border-radius`
`input:focus`	`outline`, `border-color`, `box-shadow`
`button`	`background`, `hover effects`, `transitions`
`select`	`appearance: none` for custom styling

Visualization:

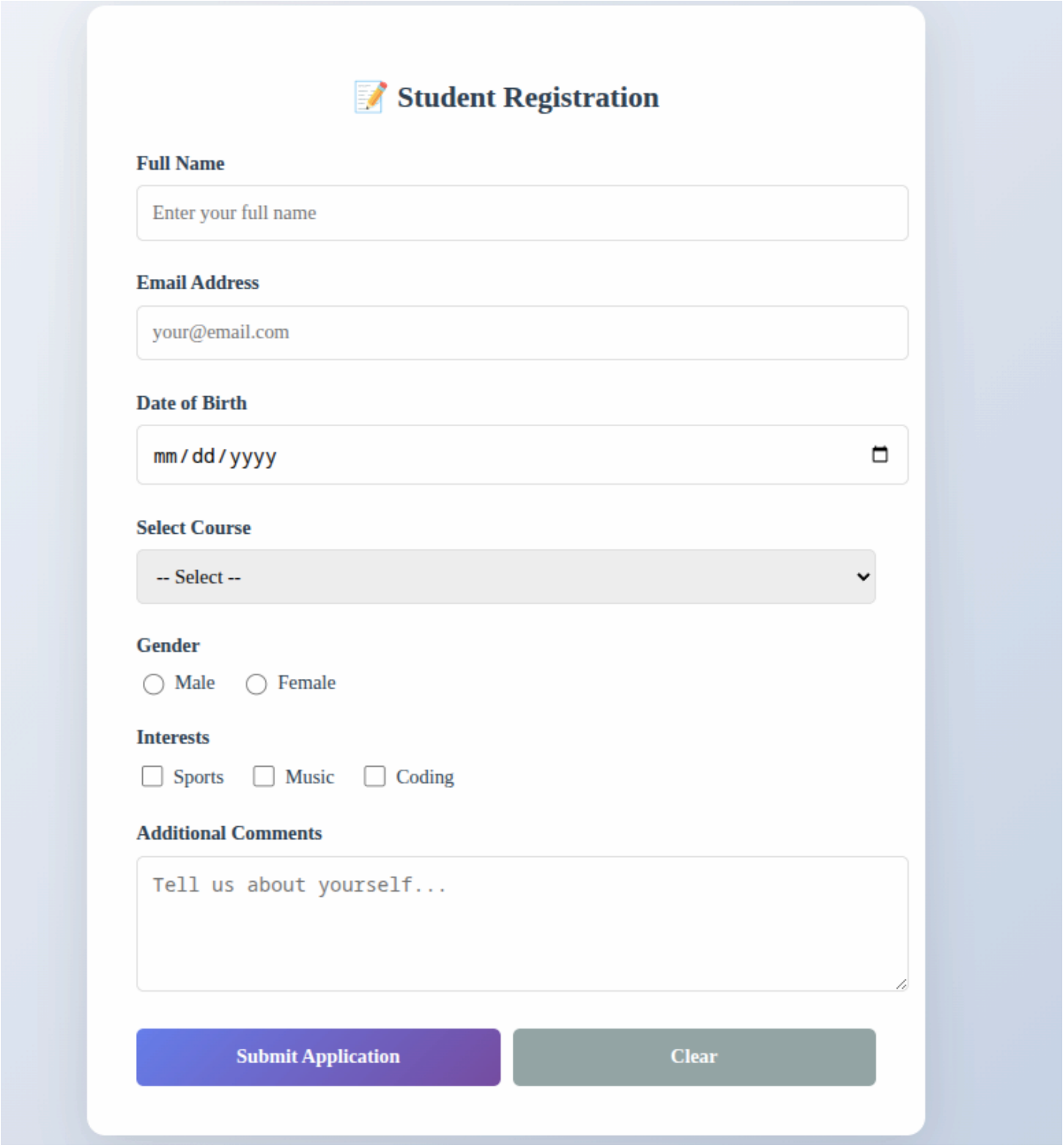
```
<!DOCTYPE html>
<html lang="en">
<head>
 <meta charset="UTF-8">
 <title>Web Forms & Controls</title>
 <style>
 body {
 font-family: 'Segoe UI';
 background: linear-gradient(135deg, #f5f7fa 0%, #c3cfe2 100%);
 padding: 40px;
 }
 .form-container {
 max-width: 600px;
 margin: 0 auto;
 background: white;
 padding: 40px;
 border-radius: 15px;
 box-shadow: 0 10px 40px rgba(0,0,0,0.1);
 }

 /* Form Group */
 .form-group {
 margin-bottom: 25px;
 }
 label {
 display: block;
 margin-bottom: 8px;
 font-weight: 600;
 color: #2c3e50;
 }

 /* Input Styling */
 input[type="text"],
 input[type="email"],
 input[type="date"],
 select,
```

```
textarea {
    width: 100%;
    padding: 12px;
    border: 2px solid #e0e0e0;
    border-radius: 6px;
    font-size: 16px;
    transition: all 0.3s;
}
input:focus, select:focus, textarea:focus {
    outline: none;
    border-color: #667eea;
    box-shadow: 0 0 0 3px rgba(102, 126, 234, 0.1);
}

/* Radio & Checkbox */
.radio-group, .checkbox-group {
    display: flex;
    gap: 20px;
    margin-top: 10px;
}
.radio-item, .checkbox-item {
    display: flex;
    align-items: center;
    cursor: pointer;
}
input[type="radio"], input[type="checkbox"] {
    width: 18px;
    height: 18px;
    margin-right: 8px;
    cursor: pointer;
}

/* Button */
button {
    width: 100%;
    padding: 14px;
    background: linear-gradient(135deg, #667eea 0%, #764ba2 100%);
    color: white;
    border: none;
    border-radius: 6px;
    font-size: 16px;
    font-weight: 600;
    cursor: pointer;
    transition: transform 0.2s;
}
button:hover {
    transform: translateY(-2px);
    box-shadow: 0 5px 15px rgba(102, 126, 234, 0.4);
```

```
        }
    </style>
</head>
<body>
    <div class="form-container">
        <h2 style="text-align: center; color: #2c3e50; margin-bottom: 30px;">     Student Registration</h2>

        <form>
            <!-- Text Input -->
            <div class="form-group">
                <label for="fullname">Full Name</label>
                <input type="text" id="fullname" placeholder="Enter your full name">
            </div>

            <!-- Email Input -->
            <div class="form-group">
                <label for="email">Email Address</label>
                <input type="email" id="email" placeholder="your@email.com">
            </div>

            <!-- Date Input -->
            <div class="form-group">
                <label for="dob">Date of Birth</label>
                <input type="date" id="dob">
            </div>

            <!-- Dropdown/Select -->
            <div class="form-group">
                <label for="course">Select Course</label>
                <select id="course">
                    <option value="">-- Select --</option>
                    <option value="bsc-cs">B.Sc. Computer Science</option>
                    <option value="ba-eco">B.A. Economics</option>
                    <option value="bsc-math">B.Sc. Mathematics</option>
                </select>
            </div>

            <!-- Radio Buttons -->
            <div class="form-group">
                <label>Gender</label>
                <div class="radio-group">
                    <div class="radio-item">
                        <input type="radio" name="gender" id="male" value="male">
                        <label for="male" style="margin: 0; font-weight: normal;">Male</label>
                    </div>
                    <div class="radio-item">
                        <input type="radio" name="gender" id="female" value="female">
                        <label for="female" style="margin: 0; font-weight: normal;">Female</label>
```

```
        </div>
    </div>
</div>

<!-- Checkboxes -->
<div class="form-group">
    <label>Interests</label>
    <div class="checkbox-group">
        <div class="checkbox-item">
            <input type="checkbox" id="sports" value="sports">
            <label for="sports" style="margin: 0; font-weight: normal;">Sports</label>
```

```
            </div>
            <div class="checkbox-item">
                <input type="checkbox" id="music" value="music">
                <label for="music" style="margin: 0; font-weight: normal;">Music</label>
            </div>
            <div class="checkbox-item">
                <input type="checkbox" id="coding" value="coding">
                <label for="coding" style="margin: 0; font-weight: normal;">Coding</label>
            </div>
        </div>
    </div>

    <!-- Textarea -->
    <div class="form-group">
        <label for="comments">Additional Comments</label>
        <textarea id="comments" rows="4" placeholder="Tell us about yourself..."></textarea>
    </div>

    <!-- Buttons -->
    <div style="display: flex; gap: 10px;">
        <button type="submit">Submit Application</button>
        <button type="reset" style="background: #95a5a6;">Clear</button>
    </div>
  </form>
 </div>
</body>
</html>
```



UNIT 3: CSS (CASCADING STYLE SHEETS)

(Weeks 8-11: Selectors, properties, box model, layout)

3.1 Concept: CSS Selectors & Specificity

What it is: Selectors target HTML elements to apply styles. Specificity determines which styles apply when conflicts occur.

Uses: - **Precise targeting:** Style specific elements without affecting others - **Efficient styling:** Reuse styles across multiple elements - **Maintenance:** Organized code that's easy to update

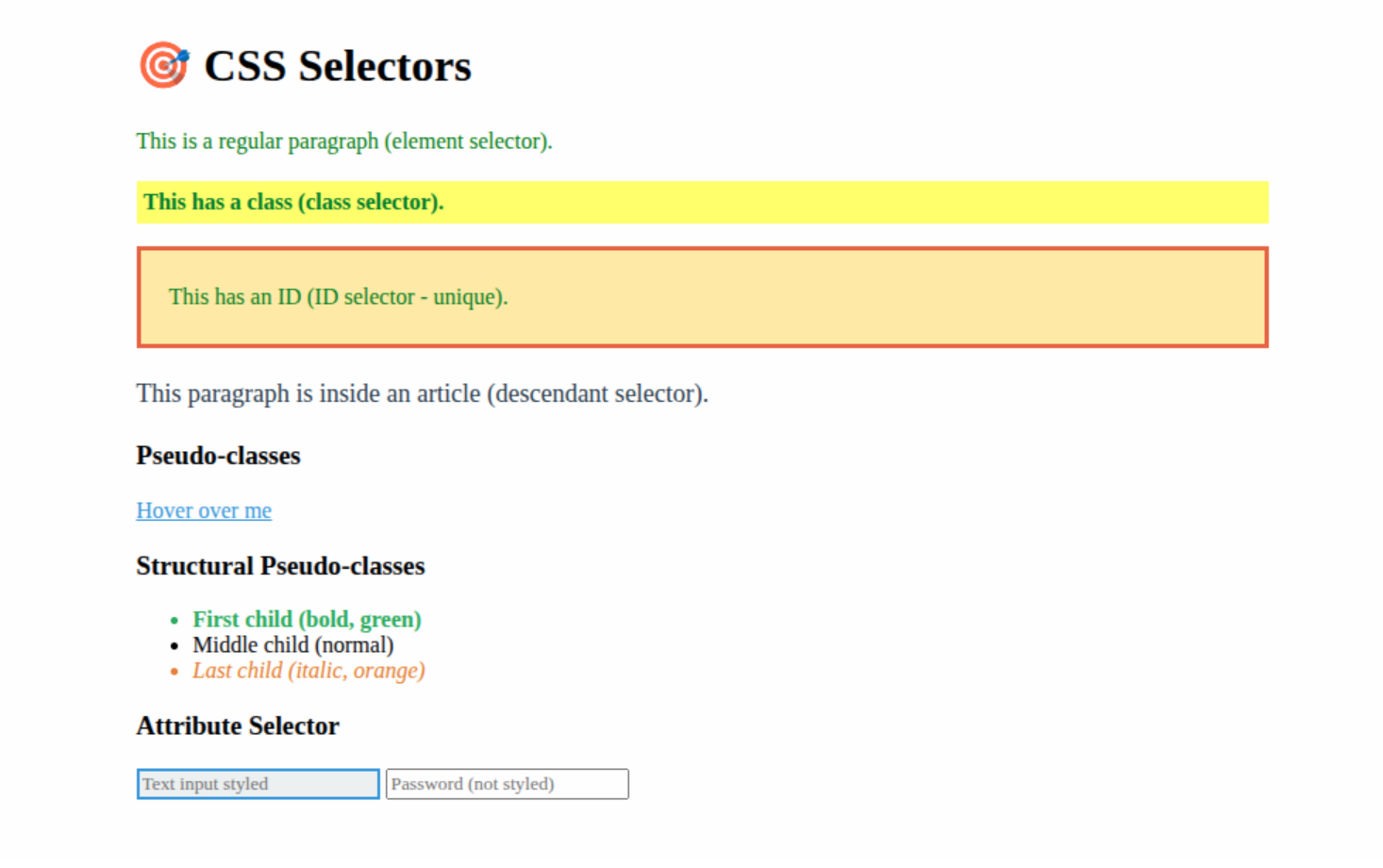
Selector Types:

Selector	Example	Targets
Element	`p`	All paragraphs
Class	`.highlight`	Elements with class="highlight"
ID	`#header`	Element with id="header"
Descendant	`nav a`	Links inside nav
Pseudo-class	`a:hover`	Links being hovered
Attribute	`input[type="text"]`	Text inputs only

Visualization:

```html
<!DOCTYPE html>
<html lang="en">
<head>
 <meta charset="UTF-8">
 <title>CSS Selectors Demo</title>
 <style>
/* Element Selector */
p {
color: #555;
line-height: 1.6;
}

/* Class Selector (Reusable) */
.highlight {
background: yellow;
padding: 2px 5px;
font-weight: bold;
}

/* ID Selector (Unique) */
#special {
border: 3px solid #e74c3c;
padding: 20px;
background: #ffeaa7;
}

/* Descendant Selector */
article p {
color: #2c3e50;
font-size: 18px;
}
```

```
        /* Pseudo-classes */
        a:link { color: #3498db; }
        a:visited { color: #9b59b6; }
        a:hover { color: #e74c3c; text-decoration: none; }
        a:active { color: #f39c12; }

        li:first-child { font-weight: bold; color: #27ae60; }
        li:last-child { font-style: italic; color: #e67e22; }

        /* Attribute Selector */
        input[type="text"] {
            border: 2px solid #3498db;
            background: #ecf0f1;
        }

        /* Specificity Demo */
        #override { color: red; }       /* ID = High specificity */
        .text { color: blue; }          /* Class = Medium */
        p { color: green; }             /* Element = Low */
    </style>
</head>
<body style="font-family: Arial; padding: 40px; max-width: 800px; margin: 0 auto;">
    <h1>     CSS Selectors</h1>

    <p>This is a regular paragraph (element selector).</p>
    <p class="highlight">This has a class (class selector).</p>
    <p id="special">This has an ID (ID selector - unique).</p>

    <article>
        <p>This paragraph is inside an article (descendant selector).</p>
    </article>

    <h3>Pseudo-classes</h3>
    <a href="#">Hover over me</a>

    <h3>Structural Pseudo-classes</h3>
    <ul>
        <li>First child (bold, green)</li>
        <li>Middle child (normal)</li>
        <li>Last child (italic, orange)</li>
    </ul>

    <h3>Attribute Selector</h3>
    <input type="text" placeholder="Text input styled">
    <input type="password" placeholder="Password (not styled)">
</body>
</html>
```



3.2 Concept: Box Model

What it is: Every HTML element is a rectangular box consisting of: Content → Padding → Border → Margin.

Uses: - **Spacing control:** Precise layout adjustments - **Visual separation:** Creating breathing room between elements - **Border effects:** Visual boundaries and decorative elements

Box Model Properties:

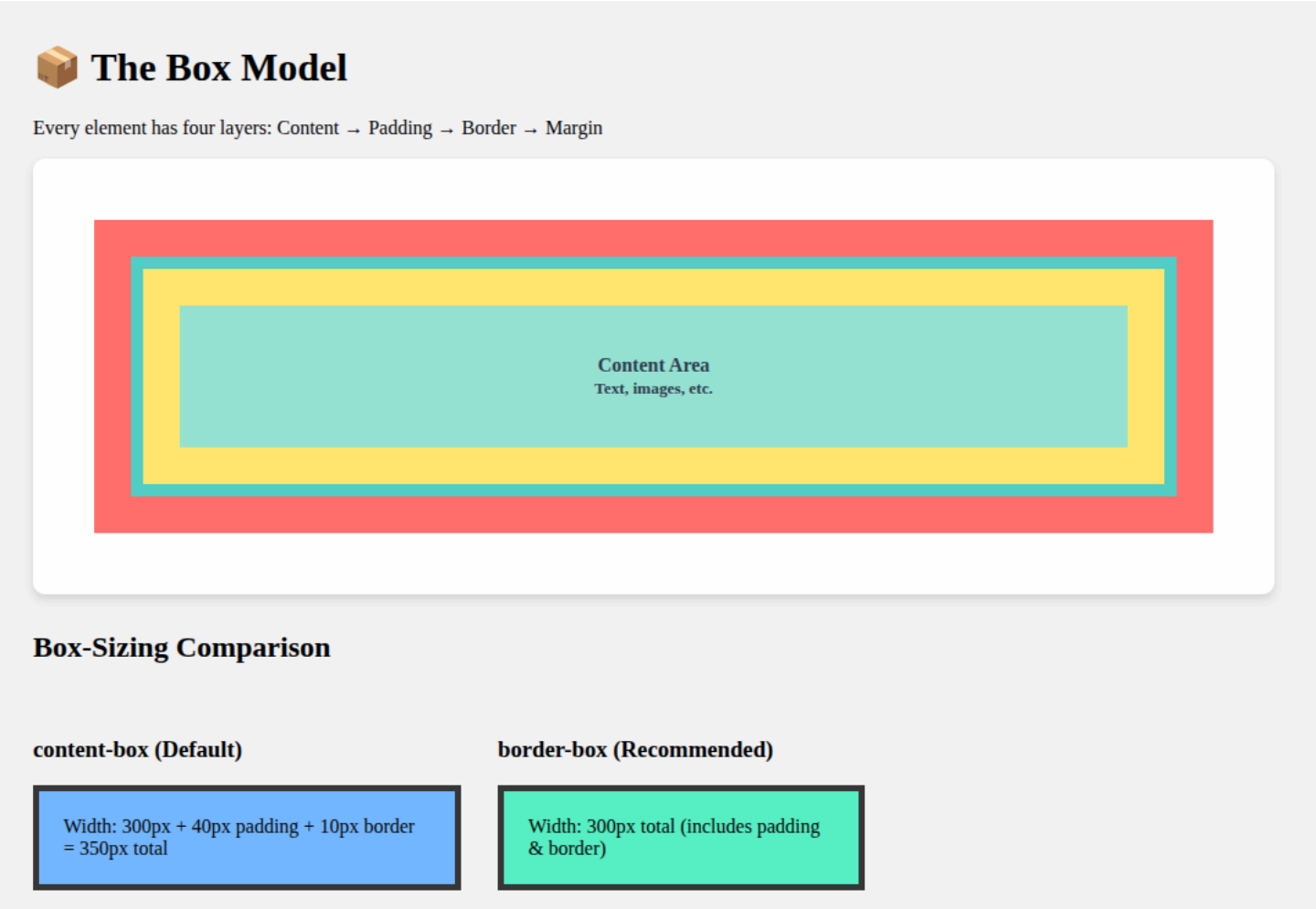
Property	Controls	Visual
`content`	Text/images	Center
`padding`	Space inside border	Around content
`border`	Edge of element	Around padding
`margin`	Space outside border	Around element

Visualization:

```html
<!DOCTYPE html>
<html lang="en">
<head>
 <meta charset="UTF-8">
 <title>Box Model Visualization</title>
 <style>
 body { font-family: Arial; background: #f0f0f0; padding: 40px; }

 /* Box Model Demonstration */
 .box-model {
 background: white;
 padding: 30px;
 border-radius: 10px;
 box-shadow: 0 4px 6px rgba(0,0,0,0.1);
 margin-bottom: 30px;
 }

 .margin-box {
 background: #ff6b6b; /* Red - Margin */
 padding: 30px;
 margin: 20px;
 }

 .border-box {
 background: #4ecdc4; /* Teal - Border */
 padding: 10px;
 }

 .padding-box {
 background: #ffe66d; /* Yellow - Padding */
 padding: 30px;
 }

 .content-box {
 background: #95e1d3; /* Green - Content */
 padding: 40px;
 text-align: center;
 font-weight: bold;
 color: #2c3e50;
 }

 /* Box Sizing Comparison */
 .container {
 display: flex;
 gap: 30px;
 margin-top: 40px;
```

```
    }

    .content-box-example {
      width: 300px;
      padding: 20px;
      border: 5px solid #333;
      background: #74b9ff;
      box-sizing: content-box; /* Default */
    }

    .border-box-example {
      width: 300px;
      padding: 20px;
      border: 5px solid #333;
      background: #55efc4;
      box-sizing: border-box; /* Preferred */
    }
  </style>
</head>
<body>
  <h1>     The Box Model</h1>
  <p>Every element has four layers: Content → Padding → Border → Margin</p>

  <div class="box-model">
    <div class="margin-box" title="Margin (outside)">
      <div class="border-box" title="Border">
        <div class="padding-box" title="Padding (inside)">
          <div class="content-box" title="Content">
            Content Area<br>
            <small>Text, images, etc.</small>
          </div>
        </div>
      </div>
    </div>
  </div>

  <h2>Box-Sizing Comparison</h2>
  <div class="container">
    <div>
      <h3>content-box (Default)</h3>
      <div class="content-box-example">
        Width: 300px + 40px padding + 10px border = 350px total
      </div>
    </div>
    <div>
      <h3>border-box (Recommended)</h3>
      <div class="border-box-example">
        Width: 300px total (includes padding & border)
```

```
        </div>
      </div>
    </div>
</body>
</html>
```



3.3 Concept: CSS Layout (Flexbox & Grid)

What it is: Modern CSS systems for creating complex layouts. Flexbox for one-dimensional layouts (rows OR columns), Grid for two-dimensional (rows AND columns).

Uses: - **Responsive design:** Adapts to screen sizes - **Centering:** Easy vertical and horizontal alignment - **Complex layouts:** Dashboards, galleries, sidebars

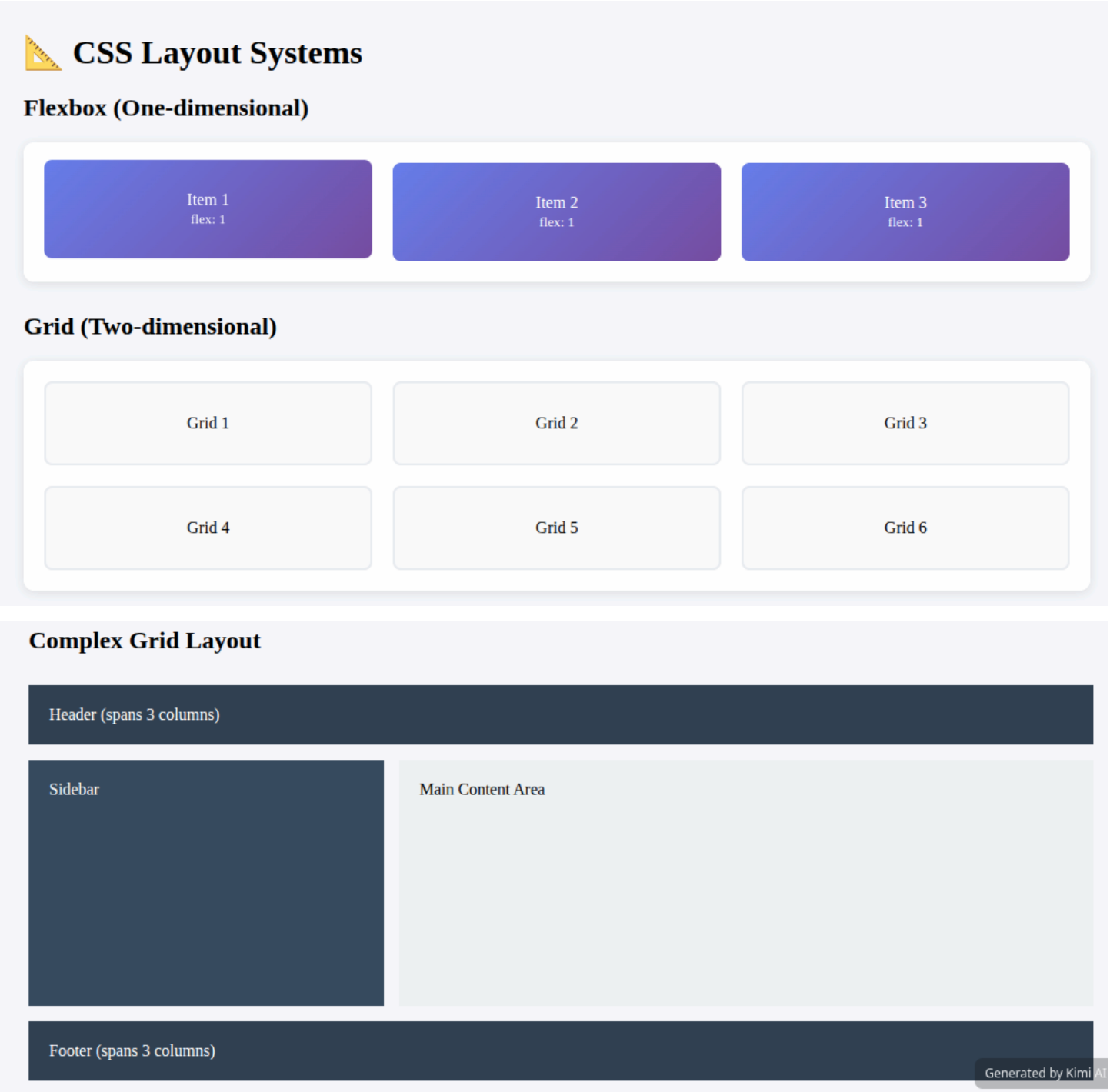
Visualization:

```
<!DOCTYPE html>
<html lang="en">
<head>
 <meta charset="UTF-8">
 <title>CSS Layout Systems</title>
```

```
<style>
  body { font-family: 'Segoe UI'; background: #f5f7fa; padding: 20px; }

  /* FLEXBOX LAYOUT */
  .flex-container {
    display: flex;
    gap: 20px;
    background: white;
    padding: 20px;
    border-radius: 10px;
    margin-bottom: 30px;
    box-shadow: 0 2px 10px rgba(0,0,0,0.1);
  }

  .flex-item {
    flex: 1; /* Equal width */
    background: linear-gradient(135deg, #667eea 0%, #764ba2 100%);
    color: white;
    padding: 30px;
    border-radius: 8px;
    text-align: center;
    transition: transform 0.3s;
  }

  .flex-item:hover {
    transform: translateY(-5px);
  }

  /* GRID LAYOUT */
  .grid-container {
    display: grid;
    grid-template-columns: repeat(3, 1fr); /* 3 equal columns */
    gap: 20px;
    background: white;
    padding: 20px;
    border-radius: 10px;
    box-shadow: 0 2px 10px rgba(0,0,0,0.1);
  }

  .grid-item {
    background: #f8f9fa;
    padding: 30px;
    border-radius: 8px;
    text-align: center;
    border: 2px solid #e9ecef;
    transition: all 0.3s;
  }
```

```
        .grid-item:hover {
            border-color: #667eea;
            background: white;
        }

        /* Complex Grid */
        .complex-grid {
            display: grid;
            grid-template-areas:
                "header header header"
                "sidebar content content"
                "footer footer footer";
            gap: 15px;
            margin-top: 30px;
        }

        .header { grid-area: header; background: #2c3e50; color: white; padding: 20px; }
        .sidebar { grid-area: sidebar; background: #34495e; color: white; padding: 20px; }
        .content { grid-area: content; background: #ecf0f1; padding: 20px; min-height: 200px; }
        .footer { grid-area: footer; background: #2c3e50; color: white; padding: 20px; }
    </style>
</head>
<body>
    <h1>     CSS Layout Systems</h1>

    <h2>Flexbox (One-dimensional)</h2>
    <div class="flex-container">
        <div class="flex-item">Item 1<br><small>flex: 1</small></div>
        <div class="flex-item">Item 2<br><small>flex: 1</small></div>
        <div class="flex-item">Item 3<br><small>flex: 1</small></div>
    </div>

    <h2>Grid (Two-dimensional)</h2>
    <div class="grid-container">
        <div class="grid-item">Grid 1</div>
        <div class="grid-item">Grid 2</div>
        <div class="grid-item">Grid 3</div>
        <div class="grid-item">Grid 4</div>
        <div class="grid-item">Grid 5</div>
        <div class="grid-item">Grid 6</div>
    </div>

    <h2>Complex Grid Layout</h2>
    <div class="complex-grid">
        <div class="header">Header (spans 3 columns)</div>
        <div class="sidebar">Sidebar</div>
        <div class="content">Main Content Area</div>
        <div class="footer">Footer (spans 3 columns)</div>
```

```
    </div>
</body>
</html>
```



UNIT 4: JAVASCRIPT

(Weeks 12-15: DOM, events, functions, validation)

4.1 Concept: DOM Manipulation

What it is: The Document Object Model (DOM) is a programming interface for web documents. JavaScript can read and modify page content dynamically.

Uses: - **Dynamic content:** Update page without reloading - **User feedback:** Show/hide elements based on interaction - **Data display:** Present calculated results

Common Methods:

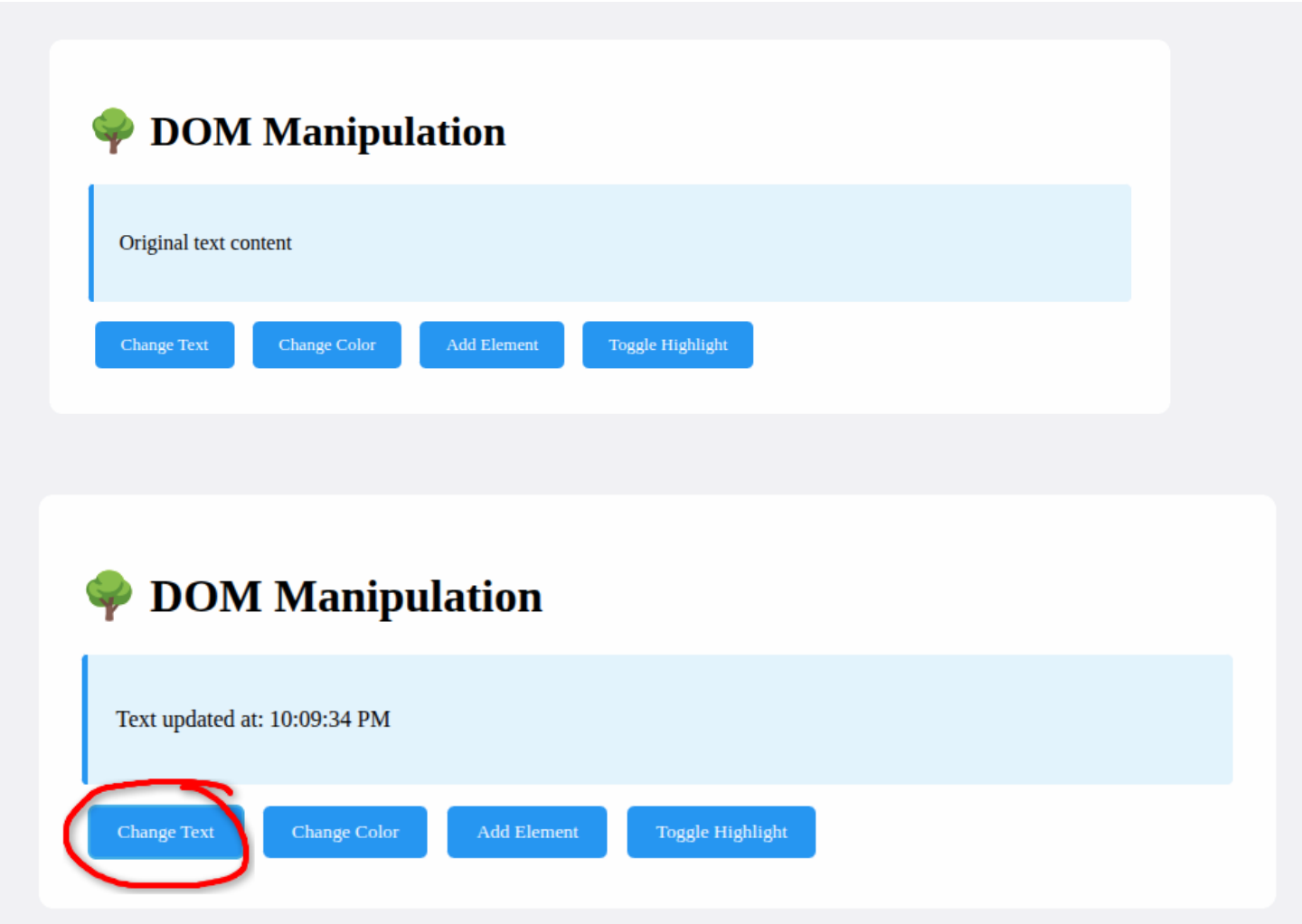
Method	Purpose
`getElementById()`	Select element by ID
`querySelector()`	Select first matching element
`querySelectorAll()`	Select all matching elements
`innerHTML`	Change HTML content
`textContent`	Change text content
`style`	Modify CSS properties

Visualization:

```html
<!DOCTYPE html>
<html lang="en">
<head>
 <meta charset="UTF-8">
 <title>DOM Manipulation</title>
 <style>
 body { font-family: 'Segoe UI'; background: #f0f2f5; padding: 40px; }
 .container { max-width: 800px; margin: 0 auto; background: white; padding: 30px; border-radius: 10px; }
 .box {
 background: #e3f2fd;
 padding: 20px;
 margin: 10px 0;
 border-left: 4px solid #2196f3;
 border-radius: 4px;
 }
 button {
 background: #2196f3;
 color: white;
 border: none;
```

```
        padding: 10px 20px;
        border-radius: 5px;
        cursor: pointer;
        margin: 5px;
    }
    button:hover { background: #1976d2; }
    .highlight { background: #fff3cd !important; border-left-color: #ffc107 !important; }
  </style>
</head>
<body>
  <div class="container">
    <h1>     DOM Manipulation</h1>

    <div id="content" class="box">
      <p id="text">Original text content</p>
    </div>

    <button onclick="changeText()">Change Text</button>
    <button onclick="changeColor()">Change Color</button>
    <button onclick="addElement()">Add Element</button>
    <button onclick="toggleHighlight()">Toggle Highlight</button>

    <div id="newElements"></div>
  </div>

  <script>
    // Change text content
    function changeText() {
      document.getElementById('text').textContent = 'Text updated at: ' + new
Date().toLocaleTimeString();
    }

    // Change style
    function changeColor() {
      document.getElementById('content').style.backgroundColor = '#e8f5e9';
      document.getElementById('content').style.borderLeftColor = '#4caf50';
    }

    // Add new element
    function addElement() {
      const newDiv = document.createElement('div');
      newDiv.className = 'box';
      newDiv.innerHTML = '<strong>New Element:</strong> Dynamically created!';
      document.getElementById('newElements').appendChild(newDiv);
    }

    // Toggle class
```

```
    function toggleHighlight() {
      document.getElementById('content').classList.toggle('highlight');
    }
  </script>
</body>
</html>
```



4.2 Concept: Event Handling

What it is: Events are actions that occur in the browser (click, keypress, load). Event handlers execute code in response.

Uses: - **Interactivity:** Respond to user actions - **Validation:** Check input before submission - **Animations:** Trigger visual effects

Common Events:

Event	Trigger
`click`	Element clicked
`submit`	Form submitted
`keydown`	Key pressed
`load`	Page finished loading
`change`	Input value changed
`mouseover`	Mouse enters element

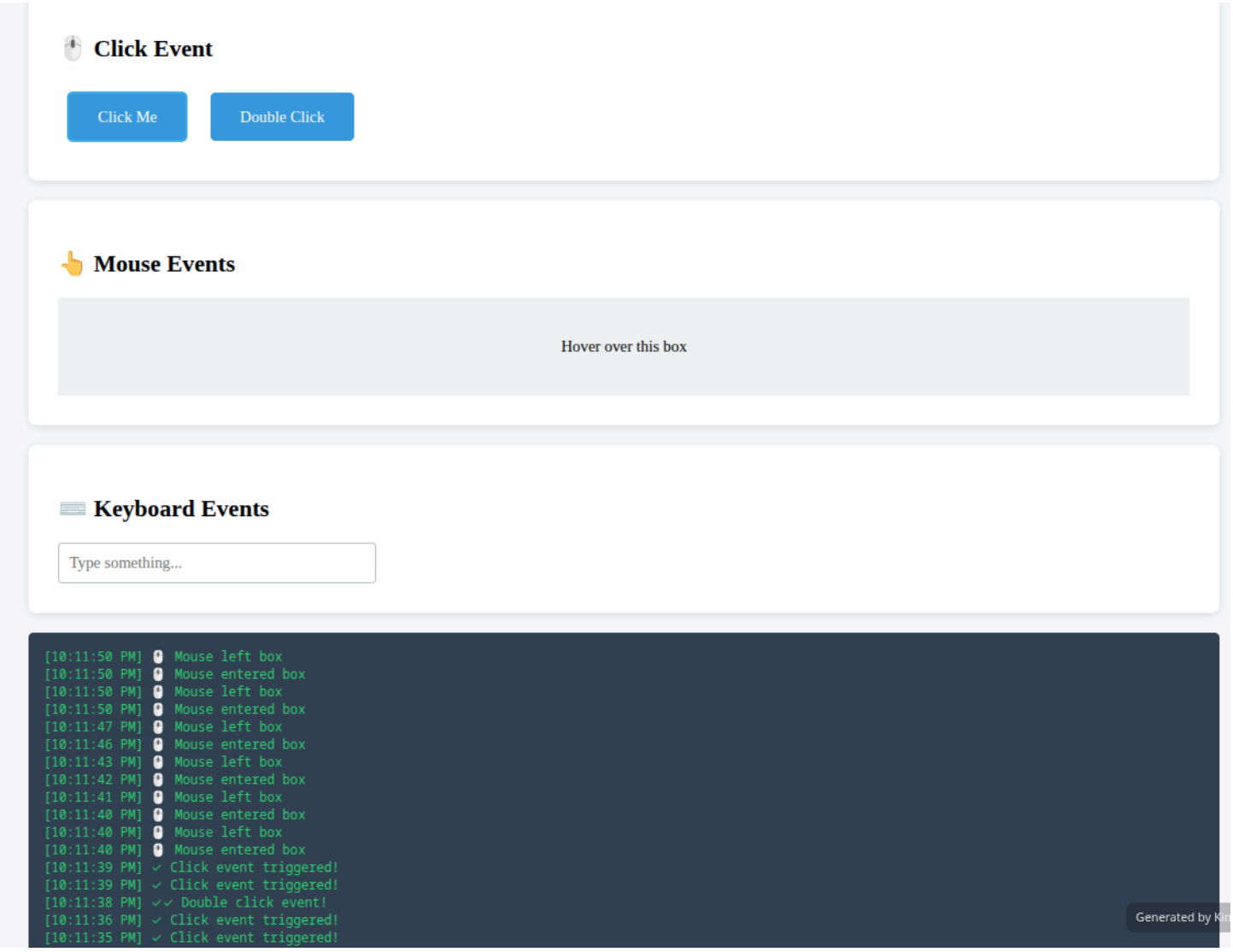
Visualization:

```html
<!DOCTYPE html>
<html lang="en">
<head>
 <meta charset="UTF-8">
 <title>Event Handling</title>
 <style>
 body { font-family: Arial; background: #f5f7fa; padding: 40px; }
 .demo-box {
 background: white;
 padding: 30px;
 margin: 20px 0;
 border-radius: 8px;
 box-shadow: 0 2px 8px rgba(0,0,0,0.1);
 }
 .event-btn {
 padding: 15px 30px;
 margin: 10px;
 border: none;
 border-radius: 5px;
 cursor: pointer;
 font-size: 16px;
 transition: all 0.3s;
 }
 .click-btn { background: #3498db; color: white; }
```

```
        .hover-box {
            background: #ecf0f1;
            padding: 40px;
            text-align: center;
            transition: all 0.3s;
        }
        .input-field {
            padding: 10px;
            font-size: 16px;
            border: 2px solid #bdc3c7;
            border-radius: 4px;
            width: 300px;
        }
        #output {
            background: #2c3e50;
            color: #2ecc71;
            padding: 15px;
            border-radius: 5px;
            font-family: monospace;
            margin-top: 20px;
            min-height: 100px;
        }
    </style>
</head>
<body>
    <div class="demo-box">
        <h2>     Click Event</h2>
        <button class="event-btn click-btn" onclick="handleClick()">Click Me</button>
        <button class="event-btn click-btn" ondblclick="handleDoubleClick()">Double Click</button>
    </div>

    <div class="demo-box">
        <h2>     Mouse Events</h2>
        <div class="hover-box" onmouseenter="mouseEnter()" onmouseleave="mouseLeave()">
            Hover over this box
        </div>
    </div>

    <div class="demo-box">
        <h2>     Keyboard Events</h2>
        <input type="text" class="input-field" placeholder="Type something..."
onkeyup="handleKeyup(event)">
    </div>

    <div id="output">Event log will appear here...</div>

    <script>
```

```
function handleClick() {
  logEvent('✓Click event triggered!');
}

function handleDoubleClick() {
  logEvent('✓✓Double click event!');
}

function mouseEnter() {
  event.target.style.background = '#d5dbdb';
  logEvent('     Mouse entered box');
}

function mouseLeave() {
  event.target.style.background = '#ecf0f1';
  logEvent('     Mouse left box');
}

function handleKeyup(e) {
  logEvent(`     Key pressed: ${e.key} (Code: ${e.keyCode})`);
}
```

```
    function logEvent(message) {
      const output = document.getElementById('output');
      output.innerHTML = `[${new Date().toLocaleTimeString()}] ${message}<br>` +
output.innerHTML;
    }
  </script>
</body>
</html>
```



4.3 Concept: Form Validation

What it is: Checking user input for correctness before submission. Can be done via HTML5 attributes or JavaScript for complex rules.

Uses: - **Data integrity:** Ensure correct format (email, phone) - **Security:** Prevent malicious input - **UX:** Immediate feedback to users

Validation Techniques:

Type	Method	Example
HTML5	`required`, `type="email"`	Basic validation
JavaScript	`if` statements, regex	Complex rules
Real-time	`onblur`, `onkeyup`	Immediate feedback

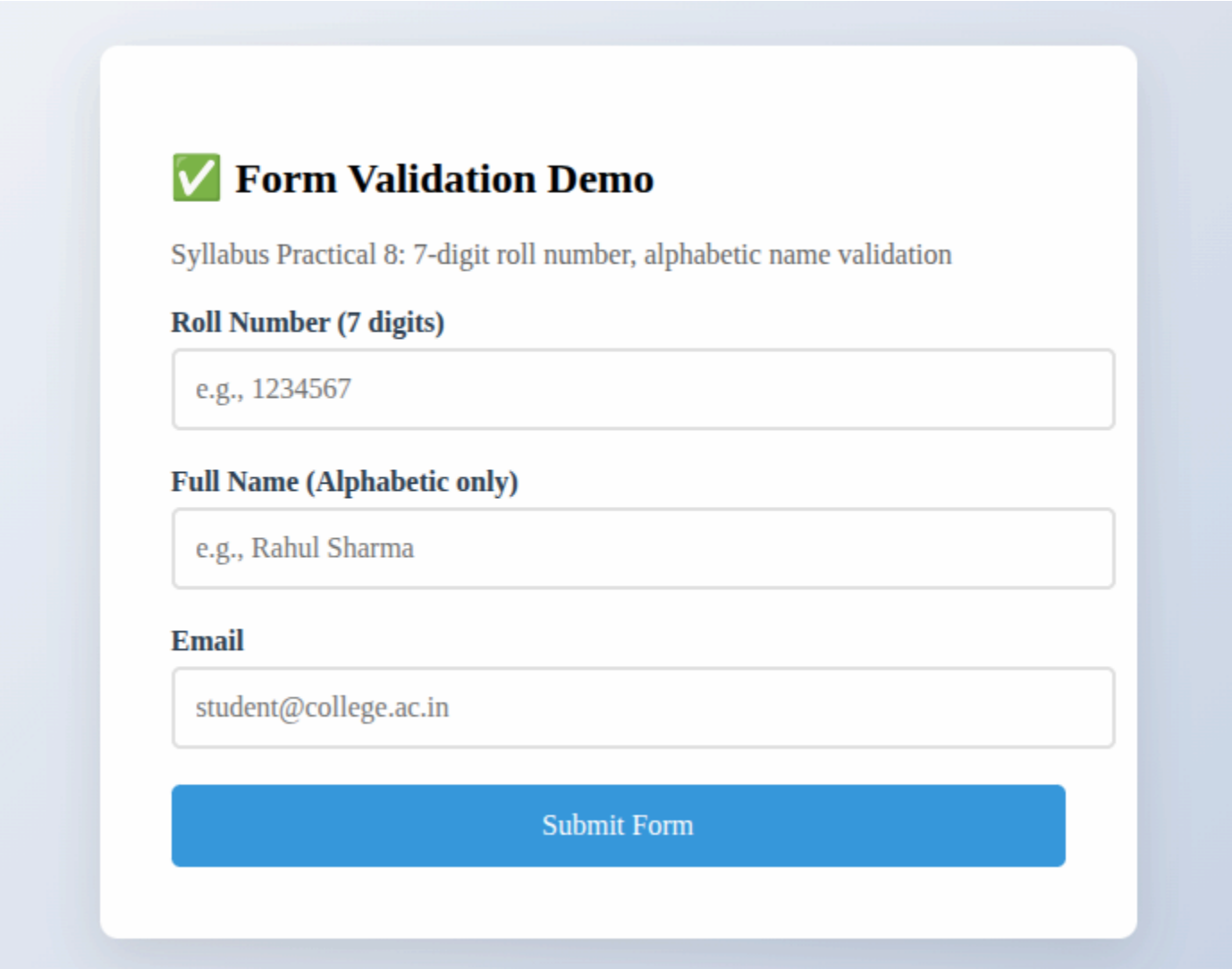
Visualization:

```
<!DOCTYPE html>
<html lang="en">
<head>
 <meta charset="UTF-8">
 <title>Form Validation</title>
 <style>
 body { font-family: 'Segoe UI'; background: linear-gradient(135deg, #f5f7fa 0%, #c3cfe2 100%); padding: 40px; }
 .form-container { max-width: 500px; margin: 0 auto; background: white; padding: 40px; border-radius: 10px; box-shadow: 0 10px 30px rgba(0,0,0,0.1); }

 .form-group { margin-bottom: 20px; }
 label { display: block; margin-bottom: 5px; font-weight: 600; color: #2c3e50; }
 input { width: 100%; padding: 12px; border: 2px solid #e0e0e0; border-radius: 5px; font-size: 16px; }
 input:focus { outline: none; border-color: #3498db; }

 input.error { border-color: #e74c3c; background: #fdf2f2; }
 input.success { border-color: #27ae60; background: #f2fdf5; }

 .error-msg { color: #e74c3c; font-size: 13px; margin-top: 5px; display: none; }
 .success-msg { color: #27ae60; font-size: 13px; margin-top: 5px; display: none; }

 button { width: 100%; padding: 14px; background: #3498db; color: white; border: none; border-radius: 5px; font-size: 16px; cursor: pointer; }
 button:hover { background: #2980b9; }

 .result { margin-top: 20px; padding: 15px; background: #d4edda; border-left: 4px solid #28a745; border-radius: 4px; display: none; }
 </style>
</head>
<body>
 <div class="form-container">
 <h2>     Form Validation Demo</h2>
 <p style="color: #666; margin-bottom: 20px;">Syllabus Practical 8: 7-digit roll number, alphabetic name validation</p>

 <form id="studentForm" onsubmit="return validateForm()">
 <div class="form-group">
```

```
            <label>Roll Number (7 digits)</label>
            <input type="text" id="rollNumber" placeholder="e.g., 1234567" onblur="validateRoll()">
            <div class="error-msg" id="rollError">     Must be exactly 7 digits</div>
            <div class="success-msg" id="rollSuccess">✓Valid roll number</div>
        </div>

        <div class="form-group">
            <label>Full Name (Alphabetic only)</label>
            <input type="text" id="fullName" placeholder="e.g., Rahul Sharma" onblur="validateName()">
            <div class="error-msg" id="nameError">     Only letters and spaces allowed</div>
            <div class="success-msg" id="nameSuccess">✓Valid name</div>
        </div>

        <div class="form-group">
            <label>Email</label>
            <input type="email" id="email" placeholder="student@college.ac.in"
onblur="validateEmail()">
            <div class="error-msg" id="emailError">     Invalid email format</div>
            <div class="success-msg" id="emailSuccess">✓Valid email</div>
        </div>

        <button type="submit">Submit Form</button>
    </form>

    <div class="result" id="successResult">
        <strong>     Registration Successful!</strong><br>
        All validations passed.
    </div>
  </div>

  <script>
    function validateRoll() {
        const roll = document.getElementById('rollNumber').value;
        const pattern = /^\d{7}$/;

        if (pattern.test(roll)) {
            showSuccess('roll');
            return true;
        } else {
            showError('roll');
            return false;
        }
    }

    function validateName() {
        const name = document.getElementById('fullName').value;
        const pattern = /^[a-zA-Z\s]+$/;
```

```
    if (pattern.test(name) && name.length >= 3) {
        showSuccess('name');
        return true;
    } else {
        showError('name');
        return false;
    }
}

function validateEmail() {
    const email = document.getElementById('email').value;
    const pattern = /^[^\s@]+@[^\s@]+\.[^\s@]+$/;

    if (pattern.test(email)) {
        showSuccess('email');
        return true;
    } else {
        showError('email');
        return false;
    }
}

function showSuccess(field) {
    document.getElementById(field + 'Number').classList.remove('error');
    document.getElementById(field + 'Number').classList.add('success');
    document.getElementById(field + 'Error').style.display = 'none';
    document.getElementById(field + 'Success').style.display = 'block';
}

function showError(field) {
    document.getElementById(field + 'Number').classList.remove('success');
    document.getElementById(field + 'Number').classList.add('error');
    document.getElementById(field + 'Error').style.display = 'block';
    document.getElementById(field + 'Success').style.display = 'none';
}

function validateForm() {
    const isRollValid = validateRoll();
    const isNameValid = validateName();
    const isEmailValid = validateEmail();

    if (isRollValid && isNameValid && isEmailValid) {
        document.getElementById('successResult').style.display = 'block';
        return false; // Prevent actual submission for demo
    }
    return false;
}
```

```
  </script>
</body>
</html>
```



SUMMARY: INTEGRATED LEARNING PATH

Week	Topic	HTML Concept	CSS Integration	JS Integration
1-3	Web Basics	Document structure	Basic styling	-
4	Text Formatting	Headings, paragraphs, formatting	Fonts, colors, backgrounds	-
5	Lists & Links	<ul>, <ol>, <a>	list-style, hover states	-

Week	Topic	HTML Concept	CSS Integration	JS Integration
6	Tables	<table>, <tr>, <td>	Borders, zebra striping	-
7	Forms & Images	<form>, <input>, <img>	Form styling, responsive images	Basic events
8-9	CSS Selectors	Classes, IDs	Specificity, cascade	-
10	Box Model	Div containers	Margin, padding, border	-
11	Layout	Semantic HTML	Flexbox, Grid	-
12-13	JavaScript Basics	-	Style manipulation	DOM, variables
14	Events	Button onclick	Visual feedback	Event handlers
15	Validation	Form attributes	Error/success states	Regex, validation logic

This structure ensures that as you learn each HTML concept, you immediately see how CSS enhances it and how JavaScript makes it interactive—following the exact syllabus while providing visual, practical understanding.